

YANGCHENG, China

WMO: 539750

Lat: 35.4888N

Lon: 112.4104E

Elev: 659

StdP: 93.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.9	-8.3	-26.0	0.4	-2.6	-23.3	0.5	-1.8	9.3	1.3	7.9	0.7	1.1	320	0.393

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	33.3	20.2	31.9	20.5	30.6	20.5	24.8	29.3	24.0	28.4	23.2	27.3	3.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	19.9	27.7	22.7	18.9	26.8	22.0	18.1	26.1	80.1	29.4	76.5	28.4	73.2	27.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.4	5.4	-12.9	36.4	2.0	1.3	-14.4	37.3	-15.5	38.0	-16.6	38.7	-18.1	39.6		
			-14.2	25.9	1.9	1.2	-15.6	26.8	-16.7	27.4	-17.7	28.1	-19.1	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-1.6	1.9	8.0	14.4	19.6	23.5	25.4	24.1	19.3	13.4	6.4	0.3	(6)
(7)		DBStd	9.85	3.06	3.68	4.08	3.92	2.93	2.51	2.23	2.66	2.98	3.10	3.60	2.92	(7)
(8)		HDD10.0	1109	359	228	87	9	0	0	0	0	8	118	300		(8)
(9)		HDD18.3	2639	618	461	319	126	21	1	0	1	22	153	359	558	(9)
(10)		CDD10.0	2191	0	0	26	142	298	406	478	438	280	114	8	0	(10)
(11)		CDD18.3	681	0	0	0	9	60	157	219	181	52	2	0	0	(11)
(12)		CDH23.3	5630	0	0	8	124	558	1435	1860	1328	296	21	0	0	(12)
(13)		CDH26.7	1916	0	0	1	29	160	552	684	423	66	1	0	0	(13)
(14)	Wind	WSAvg	1.8	1.6	1.9	2.3	2.3	2.3	2.0	1.8	1.6	1.5	1.5	1.6	1.8	(14)
(15)	Precipitation	PrecAvg	616	6	11	23	36	50	64	148	121	88	41	22	7	(15)
(16)		PrecMax	850	29	42	67	82	133	176	469	319	223	152	86	28	(16)
(17)		PrecMin	335	0	0	0	5	6	6	22	10	20	0	0	0	(17)
(18)		PrecStd	131	7	11	18	21	37	46	90	63	53	33	23	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	15.4	24.2	30.0	33.0	35.2	35.4	33.9	31.3	25.6	19.6	11.2	(19)
(20)		0.4%	MCWB	2.3	6.1	10.8	14.7	16.2	18.2	21.8	21.4	19.3	14.4	9.5	3.3	(20)
(21)		2%	DB	7.8	12.7	20.5	26.6	30.3	33.1	33.2	32.0	28.3	23.2	17.0	9.1	(21)
(22)		2%	MCWB	0.9	4.7	9.5	13.7	15.7	18.4	22.1	22.3	19.1	13.5	8.4	2.4	(22)
(23)		5%	DB	5.8	10.4	17.9	24.2	28.3	31.5	31.6	30.5	26.5	21.1	14.7	7.5	(23)
(24)		5%	MCWB	-0.1	3.6	8.3	12.7	15.6	18.2	22.4	22.1	18.1	12.6	7.5	1.4	(24)
(25)		10%	DB	4.0	8.4	15.4	21.9	26.5	30.0	30.3	29.0	24.7	19.3	12.5	5.8	(25)
(26)		10%	MCWB	-0.8	2.5	7.1	11.8	15.2	18.4	22.4	21.8	17.5	12.1	6.6	0.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.3	7.8	13.6	18.6	20.2	23.4	26.2	25.4	23.2	17.1	11.5	5.6	(27)
(28)		0.4%	MCDWB	7.5	12.5	20.2	22.7	25.9	28.8	31.5	30.1	27.3	21.1	16.0	7.7	(28)
(29)		2%	WB	1.7	5.7	10.6	16.5	18.9	22.1	25.2	24.6	21.3	15.6	9.9	3.5	(29)
(30)		2%	MCDWB	6.3	10.8	18.1	22.9	25.3	27.6	29.8	29.1	25.5	19.6	13.8	7.6	(30)
(31)		5%	WB	0.5	4.3	9.2	14.7	17.8	21.2	24.4	23.9	19.9	14.6	8.8	2.2	(31)
(32)		5%	MCDWB	5.1	9.2	16.4	20.7	24.1	26.9	28.9	28.0	23.9	18.5	12.9	6.3	(32)
(33)		10%	WB	-0.5	3.1	7.8	13.4	16.9	20.3	23.6	23.1	18.7	13.6	7.6	1.0	(33)
(34)		10%	MCDWB	3.5	7.9	14.0	19.3	23.2	26.3	27.9	27.1	22.5	17.4	11.4	5.0	(34)
(35)	Mean Daily Temperature Range		MDBR	9.5	9.9	10.9	11.4	11.5	10.8	8.5	8.2	8.8	9.6	9.6	9.0	(35)
(36)		5% DB	MCDBR	12.6	13.6	15.7	16.0	15.7	14.5	11.3	10.6	12.2	13.5	13.5	12.3	(36)
(37)			MCWBR	7.8	7.7	7.7	6.3	5.8	4.5	3.6	3.6	4.6	5.8	7.0	7.4	(37)
(38)			MCDBR	11.5	12.1	13.9	12.5	11.6	10.1	8.5	8.2	9.2	9.6	10.6	10.6	(38)
(39)		5% WB	MCWBR	7.3	7.1	7.0	5.8	5.4	4.1	3.5	3.5	4.3	4.9	5.9	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.503	0.591	0.605	0.620	0.632	0.748	0.753	0.748	0.675	0.647	0.540	0.476	(40)	
(41)		taud	1.760	1.621	1.634	1.635	1.658	1.492	1.533	1.523	1.614	1.606	1.738	1.832	(41)	
(42)		Ebn at Noon	639	628	675	696	696	618	611	601	615	575	598	634	(42)	
(43)		Edh at Noon	174	225	241	252	250	294	281	279	243	225	177	153	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.52	3.04	4.17	5.02	5.53	5.31	4.71	4.35	3.63	3.17	2.62	2.37	(44)	
(45)		RadStd	0.29	0.52	0.43	0.44	0.45	0.43	0.44	0.53	0.38	0.46	0.42	0.29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page